Erfan Sobhaei

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Education

- **M.Sc. in Computer Science**, *Sharif University of Technology, Tehran, Iran* Sep 2023 – Jan 2026
  GPA: 18.80/20
  Thesis: Overparameterization, Neural Collapse, and Their Role in Bias-Variance Alignment
  Relevant Coursework: Machine Learning Theory, Deep Learning, Statistical Learning, Randomized Algorithms.
- **B.Sc. in Computer Engineering**, *Kharazmi University, Tehran, Iran* Sep 2018 – Feb 2023
  GPA: 18.22/20
  Relevant Coursework **(GPA: 18.9/20)**: Artificial Neural Networks (graduated course), Artificial Intelligence and Expert Systems, Signals and Systems, Design of Algorithms, Engineering Probability and Statistics, Advanced Programming, Data Structures and Algorithms.

Research Interests

• Deep Learning Theory • High-Dimensional Statistics • Generalization and Robustness • Representation Learning

Research Experience

- **Graduate Research Assistant**, *Robust and Interpretable Machine Learning Lab* (RIML) Feb 2025 – Jan 2026
  - Researching group robustness and implicit regularization under supervision of Prof. Rohban and Prof. Soleymani
  - Collaborating with research teams on theoretical and experimental analysis of implicit regularization
- **Graduate Research Assistant**, *Sharif University of Technology* Oct 2024 – Dec 2025
  - Researching theoretical aspects of deep learning under supervision of Prof. Zarei
  - Studying Neural Collapse, Bias-Variance Alignment, and Canonical Representation Hypothesis and their connection

Publications

- Mirzaie, N., Alipanah, A., Abbasi, A., Farzane, A., Jafarinia, H., Sobhaei, E., Ghaznavi, M., Najafi, A., Soleymani Baghshah, M., Rohban, M. H., *Implicit Regularization of SGD Reduces Shortcut Learning*, ICLR 2026.
- Mirzaie, N., Ghaznavi, M., Oyarhoseini, H., Alipanah, A., Sobhaei, E., Abbasi, A., Farzane, A., Jafarinia, H., Sharifi Sedeh, P., Boushehrian, A., Soleymani Baghshah, M., & Rohban, M. *The Silent Helper: How Implicit Regularization Enhances Group Robustness*. In HiLD at ICML 2025 (Poster).

Experience

- **Teaching Assistant**, *Sharif University of Technology* Jun 2024 – Present
  - Machine Learning Theory, Prof. Seyyedsalehi, Spring 2026
  - Deep Learning, Prof. Soleymani, Spring 2026
  - High-Dimensional Statistics, Prof. Rohban, Fall 2025
  - Security and Privacy in Machine Learning, Prof. Sadeghzadeh, Fall 2025
  - Mathematics for Machine Learning, Prof. Najafi, Summer 2025
  - Deep Learning, Prof. Seyyedsalehi, Spring 2025
  - Machine Learning, Prof. Sharifi-Zarchi, Fall 2024
- **Artificial Intelligence Tutor**, *Darsoon* Apr 2025 – Present
- **Artificial Intelligence Tutor**, *Allameh Helli 10 High-school*, Fall 2022
- **Teaching Assistant**, *Kharazmi University* Oct 2020 – Jun 2022
  - Artificial Neural Networks, Prof. Mansouri, Spring 2022
  - Advanced Programming, Prof. Mansouri, Spring 2022
  - Artificial Intelligence and Expert Systems, Prof. Pedram, Fall 2021
  - Data Structures and Algorithms, Prof. Ashrafi-Payaman, Spring 2021, Fall 2020

Events

- 6th Youth in High-Dimensions, International Centre for Theoretical Physics, Italy 2025
- The 2nd Symposium on Frontiers in Computer and Data Sciences, Tehran Institute for Advanced Studies, Iran 2025
- ITTC-SC 29 National Mirror Committee member, Sharif University of Technology, Iran 2020
- Deep Learning Summer School, Kharazmi University, Iran 2019

Honors And Awards

- Ranked **5th GPA** among 16 graduate students of Computer Science, *Sharif University of Technology*, Apr 2025.
- Ranked **20th** place among all graduated applicants (approximately 2,000) for the Iranian University Entrance Exam in Computer Science field, Iran, 2023.
- Ranked **3rd GPA** among 50 undergraduate students of the Computer Engineering, *Kharazmi University*, Sep 2021.
- Ranked within the **top 4%** place among all applicants for the Iranian University Entrance Exam (Approximately 145,000 applicants) Iran, 2018.

Skills

- **Languages:** Persian (Native) and English (C1, TOEFL scheduled for Sept. 2026)
- **Programming and Scripting:** Python, Java, LaTeX
- **Technical Tools & Frameworks:** Scikit-learn, OpenCV, Keras, TensorFlow, PyTorch, Optuna, Wandb

Activities

- Elected Member and Head of Student's Scientific Association, Department of Electrical and Computer Engineering, *Kharazmi University*, Aug 2021 – Dec 2021.
- "Esharehgar" magazine volunteered writer and editor, Department of Electrical and Computer Engineering, *Kharazmi University*, Aug 2021 – Dec 2021.

Hobbies

- **Sports:** Soccer, table tennis, chess
- **Other:** Board games, video games, sitcoms

References

Available upon request.

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